COURSE TITLE : PRODUCTION MANAGEMENT
COURSE CODE : 5124
COURSE CATEGORY : E
PERIODS PER WEEK : 4
SEMESTER : 5
PERIODS PER SEMESTER : 52
CREDITS : 3

TIME SCHEDULE

Module	**Topics**	**Periods**
I	Introduction Production,Productivity Production Planing and Control Test	6 6 1
II	Plant location and layout Plant maintenance Material handling	6 6 1
III	Method study Work study Work measurement Value Engineering Test	6 6 1
IV	Estimating Costing Break even analysis Test	6 6 1
	Total	**52**

Course Outcome :

Module	G.O	Student will be able to:
1	1	Know Production Management
	2	Understand Production Planning And Control
2	1	Comprehend Plant Location And Layout
	2	Understand Plant Maintenance
	3	Comprehend Material Handling
3	1	Understand Method Study &Work Study
	2	Understand Work Study &Work Measurement
	3	Comprehend Value Engineering
4	1	Understand Estimating Costing

Specific outcome

MODULE-I

1.1.0 **Know Production Management**

- 1.1.1 State the scope of production management
- 1.1.2 Define production
- 1.1.3 Define productivity
- 1.1.4 Differentiate between production and productivity
- 1.1.5 Compute productivity

1.2.0 **Understand Production Planning and Control**

- 1.2.1 State PPC
- 1.2.2 Explain production planning l
- 1.2.3 Explain about PPC objectives
- 1.2.4 Explain type of production

MODULE-II

2.1.0 **Comprehend Plant Location and Layout**

- 2.1.1 Define plant location
- 2.1.2 Explain the plant selection factors
- 2.1.3 Define plant lay out
- 2.1.4 Explain objectives
- 2.1.5 Explain the factors of plant lay out
- 2.1.6 Explain different types of plant lay out

2.2.0 **Understand Plant Maintenance**

- 2.2.1 State plant maintenance
- 2.2.2 Explain type of maintenance about preventive predictive & break down
- 2.2.3 Explain about maintenance documentation

2.3.0 **Comprehend Material Handling**

- 2.3.1 State material handling
- 2.3.2 Explain the factors effecting material handling
- 2.3.4 Explain different types of material equipment

MODULE-III

3.1.0 Understand Method Study &Work Study

3.1.1 Define work study
3.1.2 State the objectives of method study
3.1.3 State the procedure of method study
3.1.4 Explain the recording techniques using charts

3.2.0 Understand Work Study &Work Measurement

3.2.1 Define work study
3.2.2 State the importance of work study
3.2.3 Explain the advantages of work study
3.2.4 Explain the procedure of work study
3.2.5 Explain work sampling

3.3.0 Comprehend Value Engineering

3.3.1 State the meaning of value engineering
3.3.2 Define value engineering
3.3.3 Differentiate between value analysis &value engineering
3.3.4 Explain the uses of value engineering
3.3.5 Explain the steps of value analyses

MODULE-IV

4.1.0 Understand Estimating Costing

4.1.1 State the objectives of estimating
4.1.2 Explain the functions of estimating
4.1.3 Explain estimation procedure
4.1.4 List different types of cost like material, labour &overheads
4.1.5 Compute costing with simple problems
4.1.6 Define depreciation
4.1.7 Calculate depreciation by straight line, reducing balance, annuity &sinking fund method
4.1.8 Define break even analysis
4.1.9 Explain the importance of break even analysis &assumptions

CONTENT DETAILS

MODULE I

1. Introduction- Explanation on scope and objective of the subject production engineering – and field of applications

2. Productivity- Introduction –definition -- Production and productivity - difference between productions - –types of production—concept of productivity—tools of productivity –productivity index–productivity measurement—factors affecting productivity – – methods of increasing productivity – - Simple problems.

3. Production planning and control a) Introduction to production planning and control--plan the work and then work the plan concept– types of production – job production batch production, mass production, continuous products – one time large production – explanation of production planning and control – benefits of PPC – functions of PPC b) Planning activities – forecasting, plant location, product planning, design and development, material selection, process planning, determination of men, machines, and material and tool requirements. Methods of Fore-casting: - Historic estimate, trend line techniques, sales force estimate, co-relation techniques, sampling techniques. Process planning – choice of machine in process planning – break even analysis – process sheet – process planning procedure c) Routing – explain routing – routing procedure – route sheet – comparison of route sheet and process sheet—comparison planning and production control d) Scheduling – factors affecting scheduling – types – master schedule parts schedule, m/c loading schedule – preparation of schedule chart in Gantt chart form e) Just in Time manufacturing – Explanation of the concept of JIT ,MRP f) Dispatching – functions – work in dispatching – list various documents prepared in dispatching g) Follow up and control – types of control material utilization control, labor utilization control, manufacturing cost control

MODULE II

1. Plant location and Plant layout a. Factors to be considered in locating Industrial plants

1. Availability of raw materials

2. proximity to market and transport facilities 3. availability of power and fuel 4. availability of suitable labor 5. Climate and atmosphere condition 5. commercial facilities 6. local laws 7. attitude of local people 8. defense from enemy attacks 9. Protected area& environmental factors. b. Plant lay out – objectives -principles of plant layout – types of layout– – product layout, process layout, combination and fixed position layout-- advantages – preparation of activity relationship chart –SLP- Application of Computers in Plant Layout- Corelap-aldep etc Factors: Influencing the plant layout --type of production – production system – availability of area – material handling system – type of building – future expansion plan-

maintenance management- Introduction – need for maintenance-cost of maintenance-direct, indirect, total cost of maintenance-unavailability cost - types of maintenance – preventive, predictive, breakdown, shutdown, scheduled maintenance – condition based maintenance – opportunity maintenance, occasional maintenance schedule – failure statistics -- maintenance log preparation – structure of maintenance department 3. Material handling Material Handling-Importance of Material Handling-Material Handling Principle-Material Handling equipments-Classification

MODULE III

1. work study a. Introduction to work study – advantages – application of work study -- work study as a productivity enhancing tool b. Introduction to method study – objectives of method study – method study procedure – state herbless and their symbols – process chart symbols – preparation of charts operation process chart, flow process chart, man-machine chart, right hand left hand chart, and silo chart – flow diagram – string diagram c. Principles of motion economy – Rules concerning Human body, work place layout and material handling, tools and equipment design

2. Work measurement a. Objectives of work measurement – procedure of stop watch time study b. Standard time calculation – production study c. Use of standard data – application – analytical estimating – P.M.T.S – M.T.M 3. Work Sampling Explain work sampling – applications – steps in work sampling – Random sampling - use of random numbers

Value Engineering Definition –uses—unnecessary cost—steps in value engineering –block diagram—product selection—fusion –analysis—system technical-- evaluation of alternatives – principle of value analysis –

MODULE-IV

Fundamentals of Estimating -Explain estimating – differentiate estimating and costing – objectives and functions of estimating – principal constituents of estimating the cost of a component – design time, method studies, time studies, drafting time, planning and production time, design and manufactures of special tools, experimental work, labor, material, over heads,Breake Even Analysis-Calculation

REFERENCE

1.Banga,Industrial organisation and Engineering Economics,S Chand

2.Martand Telsang, Production Management,S Chand